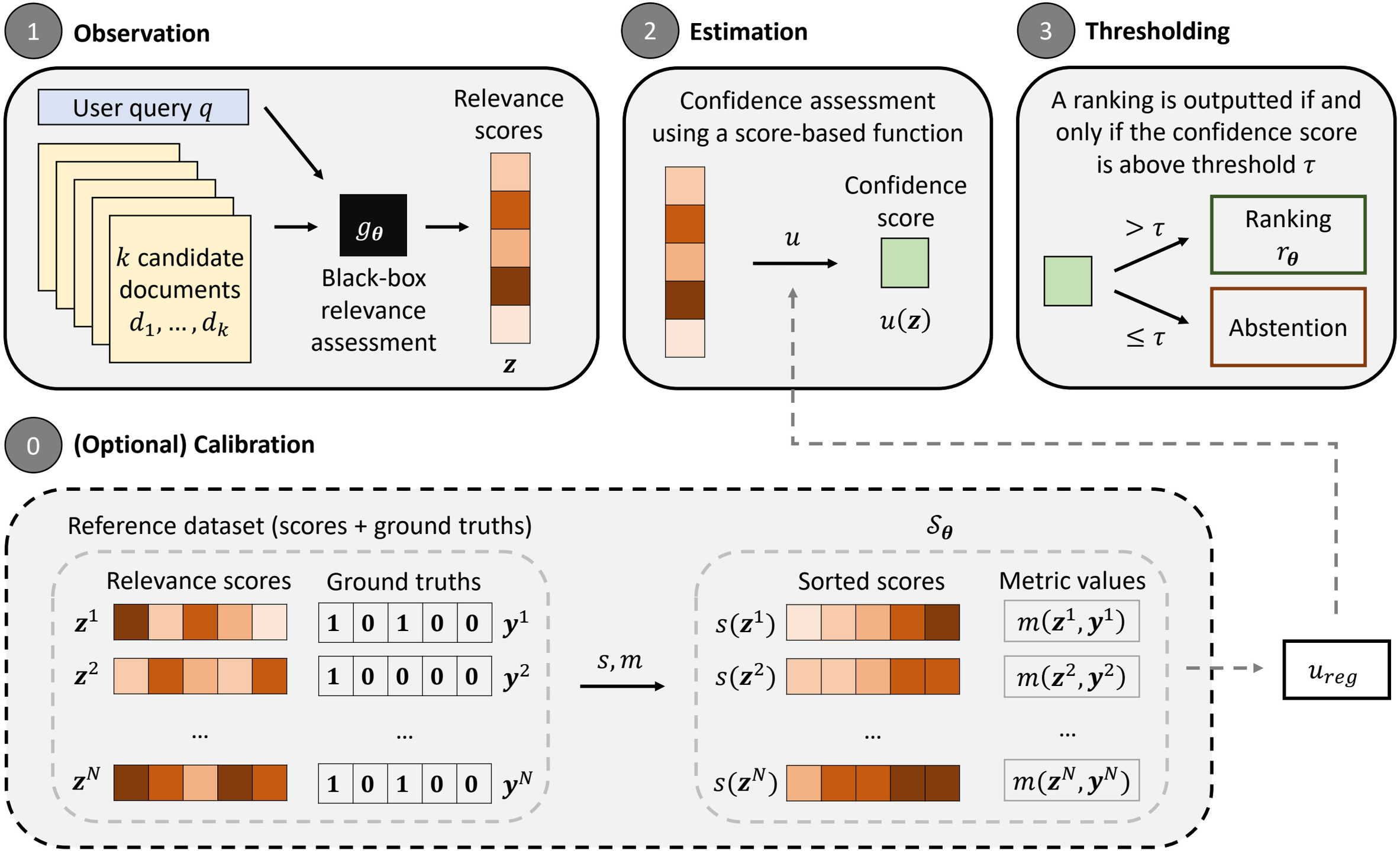

1
Observation
User query q
k candidate documents $d_1, \dots, d_k$
g_θ
Black-box relevance assessment
Relevance scores
$\boldsymbol{z}$
2
Estimation
Confidence assessment using a score-based function
u
Confidence score
$u(\boldsymbol{z})$
3
Thresholding
A ranking is outputted if and only if the confidence score is above threshold τ
$> \tau$
$\leq \tau$
Ranking r_θ
Abstention
0
(Optional) Calibration
Reference dataset (scores + ground truths)
Relevance scores
Ground truths
$\boldsymbol{z}^1$ 1 0 1 0 0 $\boldsymbol{y}^1$
$\boldsymbol{z}^2$ 1 0 0 0 0 $\boldsymbol{y}^2$
...
$\boldsymbol{z}^N$ 1 0 1 0 0 $\boldsymbol{y}^N$
s, m
$\mathcal{S}_\theta$
Sorted scores
Metric values
$s(\boldsymbol{z}^1)$ $m(\boldsymbol{z}^1, \boldsymbol{y}^1)$
$s(\boldsymbol{z}^2)$ $m(\boldsymbol{z}^2, \boldsymbol{y}^2)$
...
$s(\boldsymbol{z}^N)$ $m(\boldsymbol{z}^N, \boldsymbol{y}^N)$
u_{reg}